Applied Mathematics for Chemists
(화학수학)

Lecture 17

Matrices

Matrix (행렬)

A list of quantities

arranged in rows (행) and columns (열)

matrix elements
(행렬 원소)

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix}$$

(m by n matrix)

Matrix Algebra

- Two matrices are equal if and only if (1) they have the same numbers of rows and columns and (2) every element of one matrix is equal to the corresponding element of the other matrix
- Sum of two matrices: $\mathbf{C} = \mathbf{A} + \mathbf{B}$ if and only if (iff)

$$c_{ij} = a_{ij} + b_{ij} \text{ for every } i \text{ and } j$$

- Product of a scalar c and a matrix $\mathbf{A}$: $\mathbf{B} = c\mathbf{A}$ iff

$$b_{ij} = ca_{ij} \text{ for every } i \text{ and } j$$

Product of Two Matrices

- Matrix product $\mathbf{C} = \mathbf{AB}$ iff

$$c_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}$$

where n is the number of columns in $\mathbf{A}$ and the number of rows in $\mathbf{B}$.

(m by n matrix) × (n by p matrix) = (m by p matrix)

Properties of Matrix Multiplication

- Associative (결합법칙)

$$\mathbf{A}(\mathbf{BC}) = (\mathbf{AB})\mathbf{C}$$

- Distributive (분배법칙)

$$\mathbf{A}(\mathbf{B} + \mathbf{C}) = \mathbf{AB} + \mathbf{AC}$$

- Not necessarily commutative (교환법칙 X)

$\mathbf{AB} \neq \mathbf{BA}$ (in some cases)

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Full Bag of New Terms

1. Special matrices: identity matrix, zero matrix
2. Partner of a matrix: <u>inverse</u>, transpose, Hermitian conjugate
3. Properties of a matrix: singular, symmetric, Hermitian, orthogonal, unitary
4. Derived quantities from a matrix: trace, <u>determinant</u>

Identity Matrix (항등행렬) $\mathbf{E}$

$$\mathbf{EA} = \mathbf{AE} = \mathbf{A}$$

$$\mathbf{E} = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{bmatrix}$$

$$E_{ij} = \delta_{ij} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

Zero Matrix (영행렬) $\mathbf{0}$

$$\mathbf{0A} = \mathbf{A0} = \mathbf{0}$$

$$\mathbf{0} = \begin{bmatrix} 0 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 0 \end{bmatrix}$$

$$0_{ij} = 0$$

Inverse of Matrix (역행렬)

$$\mathbf{A}\mathbf{A}^{-1} = \mathbf{A}^{-1}\mathbf{A} = \mathbf{E}$$

- Only square matrices have inverses
- In terms of matrix elements,

$$\sum_{k=1}^{n} a_{ik}(A^{-1})_{kj} = \delta_{ij}$$

Gauss-Jordan Elimination

- Construct an augmented matrix (첨가행렬)

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} (A^{-1})_{11} & (A^{-1})_{12} \\ (A^{-1})_{21} & (A^{-1})_{22} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \rightarrow \left[\begin{array}{cc|cc} a_{11} & a_{12} & 1 & 0 \\ a_{21} & a_{22} & 0 & 1 \end{array}\right]$$

- Make an upper triangular matrix with diagonals = 1

$$\left[\begin{array}{cc|cc} 1 & * & * & * \\ 0 & 1 & * & * \end{array}\right]$$

- Then make the identity matrix

$$\left[\begin{array}{cc|cc} 1 & 0 & * & * \\ 0 & 1 & * & * \end{array}\right] \quad \mathbf{A}^{-1}$$

Transpose of Matrix (전치행렬) $\tilde{\mathbf{A}}$

Rows → columns, columns → rows

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix}$$

$$\tilde{\mathbf{A}} = \begin{bmatrix} a_{11} & a_{21} & a_{31} \\ a_{12} & a_{22} & a_{32} \end{bmatrix}$$

$$\left(\tilde{\mathbf{A}}\right)_{ij} = \tilde{a}_{ij} = a_{ji}$$

Hermitian Conjugate of Matrix $\mathbf{A}^\dagger$

(에르미트 켤레 행렬)

Transpose + complex conjugate of each element

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix}$$

$$\mathbf{A}^\dagger = \begin{bmatrix} a_{11}^* & a_{21}^* & a_{31}^* \\ a_{12}^* & a_{22}^* & a_{32}^* \end{bmatrix}$$

$$\left(\mathbf{A}^\dagger\right)_{ij} = a_{ji}^*$$

Properties of a Square Matrix

- Singular matrix (특이행렬) has no inverse

- Symmetric matrix (대칭행렬): $\mathbf{A} = \tilde{\mathbf{A}}$
- Hermitian matrix (에르미트행렬): $\mathbf{A} = \mathbf{A}^\dagger$

- Orthogonal matrix (직교행렬): $\mathbf{A}^{-1} = \tilde{\mathbf{A}}$
- Unitary matrix (유니타리행렬): $\mathbf{A}^{-1} = \mathbf{A}^\dagger$

Today's Class

- Trace and determinant
  - Trace and determinant
  - How to calculate the determinant
  - Properties of the determinant
- Applications of the determinant
  - Quantum mechanics
  - Vector product
  - Jacobian
  - Singular matrix
  - Matrix eigenvalue problem

Trace of a Square Matrix (대각합)

$$\mathrm{tr}\,\mathbf{A} = \sum_{i=1}^{n} a_{ii}$$

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \rightarrow \mathrm{tr}\,\mathbf{A} = a_{11} + a_{22} + a_{33}$$

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Property of Trace

Even when $\mathbf{AB} \neq \mathbf{BA}$,

$$\mathrm{tr}(\mathbf{AB}) = \mathrm{tr}(\mathbf{BA})$$

$$\mathrm{tr}(\mathbf{AB}) = \sum_{i=1}^{n} (\mathbf{AB})_{ii} = \sum_{i=1}^{n} \sum_{k=1}^{n} a_{ik} b_{ki}$$

$$= \sum_{k=1}^{n} \sum_{i=1}^{n} b_{ki} a_{ik} = \sum_{k=1}^{n} (\mathbf{BA})_{kk} = \mathrm{tr}(\mathbf{BA})$$

Determinant of a Square Matrix (행렬식)

$$\det \mathbf{A} = \begin{vmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \cdots & a_{nn} \end{vmatrix}$$

- 2 by 2 determinant

$$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{12}a_{21}$$

Example 13.17

Find the value of the determinant

$$\begin{bmatrix} 3 & -17 \\ 1 & 5 \end{bmatrix}.$$

Using the definition of the 2 by 2 determinant:

$$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{12}a_{21}$$

$$\begin{vmatrix} 3 & -17 \\ 1 & 5 \end{vmatrix} = 3 \times 5 - (-17) \times 1 = 32$$

Expanding the Determinant

1. Pick a row or a column (with many zeros)
2. Calculate each element × sign × minor
   - Minor: determinant after deleting the row and the column containing the element
   - Sign: counting the number of steps of one row or one column required to get from the upper left element
     → even number: (+), odd number: (-)
3. Sum them up across the row or the column
4. Repeat this up to a sum of 2 by 2 determinants

Example 13.18

Expand the 3 by 3 determinant of a general matrix $\mathbf{A}$.

$$\det \mathbf{A} = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

Pick the first row:

$$\det \mathbf{A} = a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

Now we can apply the definition of the 2 by 2 determinant:

$$\det \mathbf{A} = a_{11}(a_{22}a_{33} - a_{23}a_{32}) - a_{12}(a_{21}a_{33} - a_{23}a_{31}) + a_{13}(a_{21}a_{32} - a_{22}a_{31})$$

Row Operations and the Determinant

- If every element in any one row or in any one column is multiplied by same scalar c, the value of the determinant = c × (original determinant).

- Adding a multiple of one row to another do not change the value of the determinant.

$$\begin{vmatrix} a_{11}+ca_{21} & a_{12}+ca_{22} & a_{13}+ca_{23} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

Properties of the Determinant (1)

- If every element in any one row or in any one column is zero, the value of the determinant is zero.
- If two rows or two columns are identical, the determinant has value zero.
- If two rows or two columns are interchanged, the determinant = (-1) × (original determinant).

Properties of the Determinant (2)

- The determinant of a triangular matrix is equal to the product of the diagonal elements.

$$\begin{vmatrix} a_{11} & 0 & 0 \\ a_{21} & a_{22} & 0 \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{11}a_{22}a_{33}$$

- The determinant of a matrix and that of its transpose are equal.

$$\det \widetilde{\mathbf{A}} = \det \mathbf{A}$$

Determinant in Quantum Mechanics

An n-electron wave function $\Psi(\vec{r}_1, \vec{r}_2, \cdots, \vec{r}_n)$ must obey

1. Antisymmetric condition:

$$\Psi(\vec{r}_2, \vec{r}_1, \cdots, \vec{r}_n) = -\Psi(\vec{r}_1, \vec{r}_2, \cdots, \vec{r}_n)$$

2. Pauli exclusion principle:

$$\Psi(\vec{r}_1, \vec{r}_1, \cdots, \vec{r}_n) = 0$$

Hartree Approximation

- Multi-electron wave function ← one-electron wave function (independent electron assumption)

$$\Psi(\vec{r}_1, \vec{r}_2, \cdots, \vec{r}_n) = \psi_1(\vec{r}_1)\psi_2(\vec{r}_2)\cdots\psi_n(\vec{r}_n)$$

- The Hartree wave function does not meet the requirements (antisymmetric condition, exclusion principle)

Hartree-Fock Approximation

- Slater determinant: resolves the problem

$$\Psi(\vec{r}_1, \vec{r}_2, \cdots, \vec{r}_n) = \frac{1}{\sqrt{n!}} \begin{vmatrix} \psi_1(\vec{r}_1) & \psi_1(\vec{r}_2) & \cdots & \psi_1(\vec{r}_n) \\ \psi_2(\vec{r}_1) & \psi_2(\vec{r}_2) & \cdots & \psi_2(\vec{r}_n) \\ \vdots & \vdots & \ddots & \vdots \\ \psi_n(\vec{r}_1) & \psi_n(\vec{r}_2) & \cdots & \psi_n(\vec{r}_n) \end{vmatrix}$$

1. Antisymmetric condition:

$$\Psi(\vec{r}_2, \vec{r}_1, \cdots, \vec{r}_n) = -\Psi(\vec{r}_1, \vec{r}_2, \cdots, \vec{r}_n)$$

2. Pauli exclusion principle:

$$\Psi(\vec{r}_1, \vec{r}_1, \cdots, \vec{r}_n) = 0$$

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Other Applications of the Determinant

- Vector product of two vectors
- Jacobian calculation
- Determinant of a singular matrix
- Matrix eigenvalue problem

Vector Product of Two Vectors

The vector product (cross product) of $\vec{A}$ and $\vec{B}$ is

$$\vec{A} \times \vec{B}$$

$$= \hat{\imath}\left(A_y B_z - A_z B_y\right) + \hat{\jmath}(A_z B_x - A_x B_z) + \hat{k}\left(A_x B_y - A_y B_x\right)$$

$$= \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

Jacobian in 2 Dimensions

$$\int\int f(x,y)\,\mathrm{d}x\,\mathrm{d}y = \int\int f(\rho,\phi)\,\frac{\partial(x,y)}{\partial(\rho,\phi)}\,\mathrm{d}\rho\,\mathrm{d}\phi$$

where

$$\frac{\partial(x,y)}{\partial(\rho,\phi)} = \begin{vmatrix} \partial x/\partial\rho & \partial x/\partial\phi \\ \partial y/\partial\rho & \partial y/\partial\phi \end{vmatrix} = \left(\frac{\partial x}{\partial\rho}\right)\left(\frac{\partial y}{\partial\phi}\right) - \left(\frac{\partial x}{\partial\phi}\right)\left(\frac{\partial y}{\partial\rho}\right)$$

Jacobian in 3 Dimensions

$$\iiint f(x,y,z)\,\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z = \iiint f(r,\theta,\phi)\frac{\partial(x,y,z)}{\partial(r,\theta,\phi)}\,\mathrm{d}r\,\mathrm{d}\theta\,\mathrm{d}\phi$$

where

$$\frac{\partial(x,y,z)}{\partial(r,\theta,\phi)} = \begin{vmatrix} \partial x/\partial r & \partial x/\partial\theta & \partial x/\partial\phi \\ \partial y/\partial r & \partial y/\partial\theta & \partial y/\partial\phi \\ \partial z/\partial r & \partial z/\partial\theta & \partial z/\partial\phi \end{vmatrix}$$

Example 9.11

Obtain the Jacobian for the transformation from Cartesian coordinates to spherical polar coordinates.

$$\frac{\partial(x,y,z)}{\partial(r,\theta,\phi)} = \begin{vmatrix} \partial x/\partial r & \partial x/\partial\theta & \partial x/\partial\phi \\ \partial y/\partial r & \partial y/\partial\theta & \partial y/\partial\phi \\ \partial z/\partial r & \partial z/\partial\theta & \partial z/\partial\phi \end{vmatrix}$$

Since $x = r\sin\theta\cos\phi\,,\ y = r\sin\theta\sin\phi\,,\ z = r\cos\theta\,,$

$$\partial x/\partial r = \sin\theta\cos\phi\,,\ \partial x/\partial\theta = r\cos\theta\cos\phi\,,\ \partial x/\partial\phi = -r\sin\theta\sin\phi\,,$$

$$\partial y/\partial r = \sin\theta\sin\phi\,,\ \partial y/\partial\theta = r\cos\theta\sin\phi\,,\ \partial y/\partial\phi = r\sin\theta\cos\phi\,,$$

$$\partial z/\partial r = \cos\theta\,,\ \partial z/\partial\theta = -r\sin\theta\,,\ \partial z/\partial\phi = 0$$

Example 9.11

Obtain the Jacobian for the transformation from Cartesian coordinates to spherical polar coordinates.

Substitution leads to

$$\frac{\partial(x,y,z)}{\partial(r,\theta,\phi)} = \begin{vmatrix} \sin\theta\cos\phi & r\cos\theta\cos\phi & -r\sin\theta\sin\phi \\ \sin\theta\sin\phi & r\cos\theta\sin\phi & r\sin\theta\cos\phi \\ \cos\theta & -r\sin\theta & 0 \end{vmatrix}$$

Pick the third row:

$$\frac{\partial(x,y,z)}{\partial(r,\theta,\phi)}$$

$$= \cos\theta \begin{vmatrix} r\cos\theta\cos\phi & -r\sin\theta\sin\phi \\ r\cos\theta\sin\phi & r\sin\theta\cos\phi \end{vmatrix} + r\sin\theta \begin{vmatrix} \sin\theta\cos\phi & -r\sin\theta\sin\phi \\ \sin\theta\sin\phi & r\sin\theta\cos\phi \end{vmatrix}$$

Example 9.11

Obtain the Jacobian for the transformation from Cartesian coordinates to spherical polar coordinates.

$$\frac{\partial(x, y, z)}{\partial(r, \theta, \phi)}$$

$$= \cos\theta \begin{vmatrix} r\cos\theta\cos\phi & -r\sin\theta\sin\phi \\ r\cos\theta\sin\phi & r\sin\theta\cos\phi \end{vmatrix} + r\sin\theta \begin{vmatrix} \sin\theta\cos\phi & -r\sin\theta\sin\phi \\ \sin\theta\sin\phi & r\sin\theta\cos\phi \end{vmatrix}$$

$$= \cos\theta\,(r^2\cos\theta\sin\theta\cos^2\phi + r^2\cos\theta\sin\theta\sin^2\phi) + r\sin\theta\,(r\sin^2\theta\cos^2\phi + r\sin^2\theta\sin^2\phi)$$

$$= r^2\cos^2\theta\sin\theta\,(\cos^2\phi + \sin^2\phi) + r^2\sin^3\theta\,(\cos^2\phi + \sin^2\phi)$$

$$= r^2\sin\theta\,(\cos^2\theta + \sin^2\theta)$$

$$= r^2\sin\theta$$

Determinant of a Singular Matrix

- A matrix with no inverse is singular
- The determinant of a singular matrix has value 0

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 1/2 & 1 \end{bmatrix}$$

$$\det \mathbf{A} = 1 \times 1 - 2 \times (1/2) = 0$$

Matrix Eigenvalue Problem

For a given square matrix $\mathbf{B}$, if we have

$$\mathbf{B}\vec{x} = b\vec{x}$$

- $\vec{x}$ (column vector): eigenvector
- b (scalar): eigenvalue

We can rearrange it as

$$(\mathbf{B} - b\mathbf{E})\vec{x} = \vec{0}$$

Use of Determinant

$$(\mathbf{B} - b\mathbf{E})\vec{x} = \vec{0}$$

If $\mathbf{B} - b\mathbf{E} \neq \mathbf{0}$ and $\vec{x} \neq \vec{0}$, we have

$$\det(\mathbf{B} - b\mathbf{E}) = 0$$

By solving this equation, one can determine b and consequently $\vec{x}$.

Example 14.6

Find the values of b and $\vec{x}$ that satisfy the eigenvalue equation

$$\begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} \vec{x} = b\vec{x}.$$

Using the determinant condition, $\det(\mathbf{B} - b\mathbf{E}) = 0$,

$$\begin{vmatrix} 1-b & 1 & 0 \\ 1 & 1-b & 1 \\ 0 & 1 & 1-b \end{vmatrix} = 0$$

Calculate the determinant:

$$(1-b)\begin{vmatrix} 1-b & 1 \\ 1 & 1-b \end{vmatrix} - \begin{vmatrix} 1 & 1 \\ 0 & 1-b \end{vmatrix} = (1-b)[(1-b)^2 - 1] - (1-b) = 0$$

Example 14.6

Find the values of b and $\vec{x}$ that satisfy the eigenvalue equation

$$\begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} \vec{x} = b\vec{x}.$$

$$(1-b)[(1-b)^2 - 2] = 0$$

Solving this equation, we obtain the eigenvalues

$$b = 1 \text{ or } b = 1 \pm \sqrt{2}$$

Now determine the eigenvectors by letting

$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Example 14.6

Find the values of b and $\vec{x}$ that satisfy the eigenvalue equation

$$\begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} \vec{x} = b\vec{x}.$$

1) $b = 1$:

$$\begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$\begin{cases} x_1 + x_2 = x_1 \\ x_1 + x_2 + x_3 = x_2 \\ x_2 + x_3 = x_3 \end{cases}$$

Hence, $x_2 = 0$ and $x_3 = -x_1$

Example 14.6

Find the values of b and $\vec{x}$ that satisfy the eigenvalue equation

$$\begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} \vec{x} = b\vec{x}.$$

1) $b = 1$: $x_2 = 0$ and $x_3 = -x_1$

The eigenvector corresponding to $b = 1$ is

$$\vec{x}(b=1) = \begin{bmatrix} x_1 \\ 0 \\ -x_1 \end{bmatrix} = C \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

(C: arbitrary constant)

Example 14.6

Find the values of b and $\vec{x}$ that satisfy the eigenvalue equation

$$\begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} \vec{x} = b\vec{x}.$$

2) $b = 1 + \sqrt{2}$:

$$\begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \left(1 + \sqrt{2}\right) \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$\begin{cases} x_1 + x_2 = \left(1 + \sqrt{2}\right) x_1 \\ x_1 + x_2 + x_3 = \left(1 + \sqrt{2}\right) x_2 \\ x_2 + x_3 = \left(1 + \sqrt{2}\right) x_3 \end{cases}$$

Example 14.6

Find the values of b and $\vec{x}$ that satisfy the eigenvalue equation

$$\begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} \vec{x} = b\vec{x}.$$

2) $b = 1 + \sqrt{2}$:

$$\begin{cases} x_2 = \sqrt{2}x_1 \\ x_1 + x_3 = \sqrt{2}x_2 \\ x_2 = \sqrt{2}x_3 \end{cases}$$

Hence, $x_2 = \sqrt{2}x_1$ and $x_3 = x_1$.

Example 14.6

Find the values of b and $\vec{x}$ that satisfy the eigenvalue equation

$$\begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} \vec{x} = b\vec{x}.$$

2) $b = 1 + \sqrt{2}$: $x_2 = \sqrt{2}x_1$ and $x_3 = x_1$

The eigenvector corresponding to $b = 1 + \sqrt{2}$ is

$$\vec{x}\left(b = 1 + \sqrt{2}\right) = \begin{bmatrix} x_1 \\ \sqrt{2}x_1 \\ x_1 \end{bmatrix} = C \begin{bmatrix} 1 \\ \sqrt{2} \\ 1 \end{bmatrix}$$

(C: arbitrary constant)

Example 14.6

Find the values of b and $\vec{x}$ that satisfy the eigenvalue equation

$$\begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} \vec{x} = b\vec{x}.$$

3) $b = 1 - \sqrt{2}$:

$$\begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \left(1 - \sqrt{2}\right) \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$\begin{cases} x_1 + x_2 = \left(1 - \sqrt{2}\right) x_1 \\ x_1 + x_2 + x_3 = \left(1 - \sqrt{2}\right) x_2 \\ x_2 + x_3 = \left(1 - \sqrt{2}\right) x_3 \end{cases}$$

Example 14.6

Find the values of b and $\vec{x}$ that satisfy the eigenvalue equation

$$\begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} \vec{x} = b\vec{x}.$$

3) $b = 1 - \sqrt{2}$:

$$\begin{cases} x_2 = -\sqrt{2}x_1 \\ x_1 + x_3 = -\sqrt{2}x_2 \\ x_2 = -\sqrt{2}x_3 \end{cases}$$

Hence, $x_2 = -\sqrt{2}x_1$ and $x_3 = x_1$.

Example 14.6

Find the values of b and $\vec{x}$ that satisfy the eigenvalue equation

$$\begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} \vec{x} = b\vec{x}.$$

3) $b = 1 - \sqrt{2}$: $x_2 = -\sqrt{2}x_1$ and $x_3 = x_1$

The eigenvector corresponding to $b = 1 - \sqrt{2}$ is

$$\vec{x}\left(b = 1 - \sqrt{2}\right) = \begin{bmatrix} x_1 \\ -\sqrt{2}x_1 \\ x_1 \end{bmatrix} = C \begin{bmatrix} 1 \\ -\sqrt{2} \\ 1 \end{bmatrix}$$

(C: arbitrary constant)